11th grade
Learning evidence 2.1: Confidence Interval Planning (Practice Solutions)

Problems

Problem 1:

C1

Using grouped midpoints x with frequencies f:

$$\sum fx = 10(20) + 16(25) + 19(30) + 17(35) + 13(40) = 2285$$

$$\bar{x} = \frac{\sum fx}{n} = \frac{2285}{75} \approx 30{,}47$$

$$\sum f(x-\bar{x})^2 \approx 10(20-30{,}47)^2 + 16(25-30{,}47)^2 + 19(30-30{,}47)^2 + 17(35-30{,}47)^2 + 13(40-30{,}47)^2$$

$$\approx 1095{,}51 + 478{,}15 + 4{,}14 + 349{,}37 + 1181{,}50 = 3108{,}67$$

$$s = \sqrt{\frac{\sum f(x-\bar{x})^2}{n-1}} = \sqrt{\frac{3108{,}67}{74}} \approx 6{,}48$$

C2

The sample statistics are $\bar{x} \approx 30{,}47$ and $s \approx 6{,}48$, while the population parameters are μ (the true mean) and $\sigma = 6{,}0$ minutes. The sample values estimate, but do not equal, the population values.

C3

A single point estimate like $\bar{x}$ ignores sampling variability. A confidence interval gives a range of plausible values for μ, which is more informative for planning in a real setting.

C5

With known $\sigma = 6{,}0$ and $n = 75$, the standard error is

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{6{,}0}{\sqrt{75}} \approx 0{,}69.$$

For 95 % confidence, $z_{0{,}025} = 1{,}96$:

$$E = 1{,}96(0{,}69) \approx 1{,}36, \quad \mu \in 30{,}47 \pm 1{,}36 = [29{,}11, 31{,}82].$$

For 98 % confidence, $z_{0{,}01} = 2{,}33$:

$$E = 2{,}33(0{,}69) \approx 1{,}61, \quad \mu \in 30{,}47 \pm 1{,}61 = [28{,}85, 32{,}08].$$

C4

We are 95 % confident the true mean loading time is between about 29.11 and 31.82 minutes, and 98 % confident it is between about 28.85 and 32.08 minutes. The 98 % interval is wider because higher confidence requires a larger margin of error.

Problem 2: **C1**

$$\sum x = 850, \quad \bar{x} = \frac{850}{20} = 42{,}5$$
$$\sum (x-\bar{x})^2 = 185, \quad s = \sqrt{\frac{185}{19}} \approx 3{,}12$$

C2

The sample statistics are $\bar{x} = 42{,}5$ and $s \approx 3{,}12$, while the population mean μ and population standard deviation $\sigma = 5{,}5$ are parameters for all lunch periods. The sample measures are estimates of the population values.

C3

Because only 20 lunch periods are observed, a point estimate may be off. Confidence intervals quantify that uncertainty and show a plausible range for μ instead of a single guess.

C5

With known $\sigma = 5{,}5$ and $n = 20$,

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{5{,}5}{\sqrt{20}} \approx 1{,}23.$$

For 95 % confidence, $z_{0{,}025} = 1{,}96$:

$$E = 1{,}96(1{,}23) \approx 2{,}41, \quad \mu \in 42{,}5 \pm 2{,}41 = [40{,}09, 44{,}91].$$

For 98 % confidence, $z_{0{,}01} = 2{,}33$:

$$E = 2{,}33(1{,}23) \approx 2{,}87, \quad \mu \in 42{,}5 \pm 2{,}87 = [39{,}63, 45{,}37].$$

C4

We are 95 % confident the true mean sandwiches sold per lunch period is between about 40.09 and 44.91, and 98 % confident it is between about 39.63 and 45.37. The wider 98 % interval offers more assurance at the cost of precision.